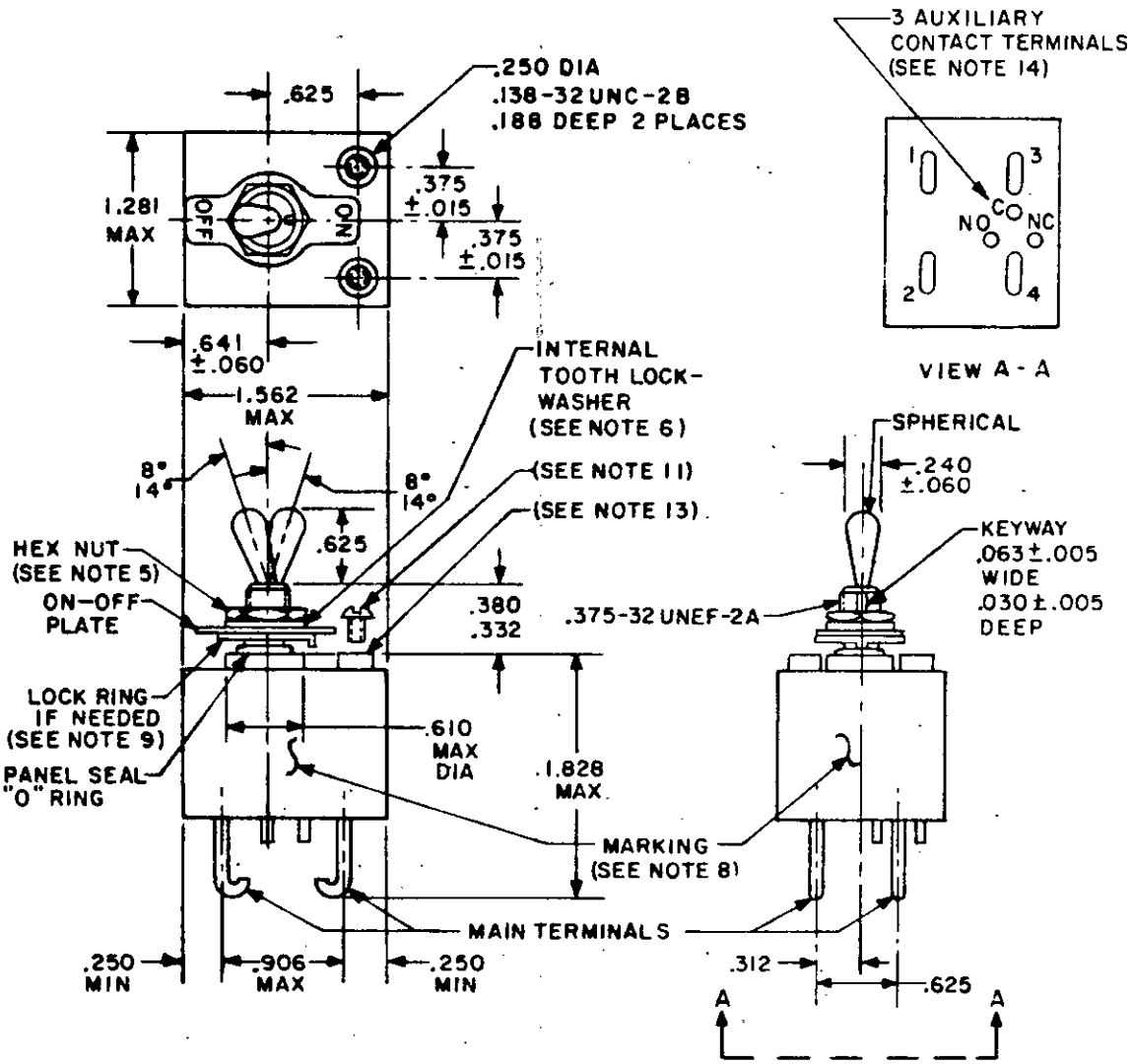
MILITARY SPECIFICATION SHEET

CIRCUIT BREAKERS, MAGNETIC, LOW-POWER, SEALED, TRIP-FREE

TWO-POLE, AUXILIARY CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the circuit breakers described herein shall consist of this specification and the latest issue of MIL-C-39019.

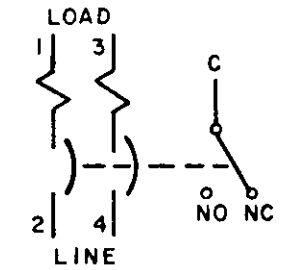


Ⓔ FIGURE 1. Circuit breaker configuration.

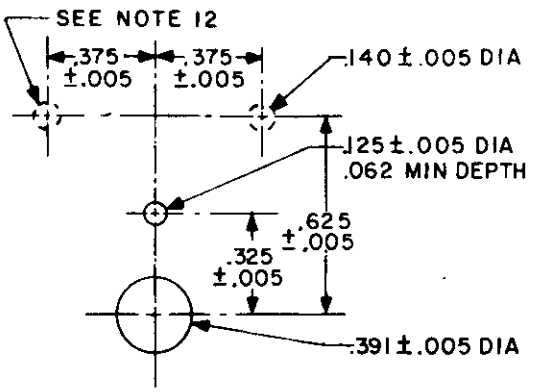
Ⓔ denotes changes

FSC 5925

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.



CIRCUIT DIAGRAM
(MAIN CONTACTS OPEN)
(SEE NOTE 10)



PANEL MTG DETAIL

INCHES	MM	INCHES	MM
.005	.13	.332	8.43
.010	.25	.375	9.52
.015	.38	.380	9.65
.025	.64	.382	9.70
.030	.76	.391	9.93
.031	.79	.392	9.96
.060	1.52	.493	12.52
.063	1.60	.500	12.70
.093	2.36	.507	12.88
.125	3.18	.610	15.49
.140	3.56	.625	15.88
.188	4.78	.641	16.28
.240	6.10	.906	23.01
.250	6.35	1.281	32.54
.312	7.92	1.562	39.67
.315	8.26	1.828	46.43

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Tolerance is ±.031 unless otherwise specified.
4. Envelope design optional.
5. Hex mounting nut .375-32 UNEF-2B thread, .500 ±.010 across flats, .093 ±.005 thick, brass nickel plated, QQ-N-290 nonglare, or stainless steel.
6. Internal tooth lockwasher, .507/.493 O.D., .392/.382 I.D., .025 ±.005 thick, stainless steel.
7. The effective bushing thread length (not including mounting hardware) is .280 minimum.
8. Marking may appear on any surface except the mounting surface.
9. Lock ring not required if on-off plate has locking tab.
10. Numerical marking optional on circuit diagram.
11. Sealscrew, .138-32 UNC-2A x .312 long, stainless steel No. 18-8 (type 347) with O-Ring-Buna N Durometer 60-70. Possible source, APM Hexseal Corp.
12. Use of panel screws will enhance circuit breaker shock and vibration characteristics.
13. Threaded inserts exceed height of bushing by .005 to .015.
14. Auxiliary contact terminals shall be located on the back side of the circuit breaker (same surface as main terminals), but otherwise their location, style and design shall be optional, consistent with dimensional or other specified requirements. Auxiliary contact terminals shall accomodate two No. 20 AWG wires.

FIGURE 1. Circuit breaker configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: For reference only, approximately 100 grams.

Mounting hardware: See figure 1.

Terminals: See figure 1.

Voltage and frequency rating: 50 Vdc, maximum and 240 Vac, maximum, at 60 and 400 Hz.

High inrush: 800 percent at 60 Hz and dc, 1400 percent at 400 Hz (see table I).

Current rating: See table I.

Circuit diagram: See figure 1.

Time delay: See table I.

Resistance or impedance: See table I.

Part number: M39019/04-(dash number taken from table I) (e.g., M39019/04-200).

Cross reference: See table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.

TABLE I. Circuit breaker dash numbers and applicable chacteristics.

Dash number		Current rating (amps)	Time delay	Resistance or impedance (ohms - max) 1/		
Standard delay	High inrush			DC	60 Hz	400 Hz
200	300	0.05	A	680.0	690.0	710.0
201	301	0.05	B	680.0	690.0	710.0
202	---	0.05	C	680.0	690.0	710.0
203	302	0.1	A	150.0	170.0	180.0
204	303	0.1	B	150.0	170.0	180.0
205	---	0.1	C	150.0	170.0	180.0
206	304	0.25	A	20.0	26.0	27.0
207	305	0.25	B	20.0	26.0	27.0
208	---	0.25	C	20.0	26.0	27.0
209	306	0.5	A	5.4	6.0	6.6
210	307	0.5	B	5.4	6.0	6.6
211	---	0.5	C	5.4	6.0	6.6
212	308	0.75	A	2.5	2.7	2.8
213	309	0.75	B	2.5	2.7	2.8
214	---	0.75	C	2.5	2.7	2.8
215	310	1.0	A	1.35	1.5	1.61
216	311	1.0	B	1.35	1.5	1.61
217	---	1.0	C	1.35	1.5	1.61
218	312	2.0	A	0.40	0.40	0.50
219	313	2.0	B	0.40	0.40	0.50
220	---	2.0	C	0.40	0.40	0.50
221	314	2.5	A	0.25	0.25	0.27
222	315	2.5	B	0.25	0.25	0.27
223	---	2.5	C	0.25	0.25	0.27
224	316	3.0	A	0.15	0.15	0.17
225	317	3.0	B	0.15	0.15	0.17
226	---	3.0	C	0.15	0.15	0.17
227	318	4.0	A	0.10	0.10	0.12
228	319	4.0	B	0.10	0.10	0.12
229	---	4.0	C	0.10	0.10	0.12
230	320	5.0	A	0.061	0.063	0.072
231	321	5.0	B	0.061	0.063	0.072
232	---	5.0	C	0.061	0.063	0.072
233	322	6.0	A	0.042	0.043	0.050
234	323	6.0	B	0.042	0.043	0.050
235	---	6.0	C	0.042	0.043	0.050
236	324	7.0	A	0.036	0.036	0.040
237	325	7.0	B	0.036	0.036	0.040
238	---	7.0	C	0.036	0.036	0.040
239	326	7.5	A	0.031	0.031	0.038
240	327	7.5	B	0.031	0.031	0.038
241	---	7.5	C	0.031	0.031	0.038
242	328	8.0	A	0.027	0.028	0.035
243	329	8.0	B	0.027	0.028	0.035
244	---	8.0	C	0.027	0.028	0.035
245	330	9.0	A	0.022	0.022	0.028
246	331	9.0	B	0.022	0.022	0.028
247	---	9.0	C	0.022	0.022	0.028
248	332	10.0	A	0.018	0.021	0.024
249	333	10.0	B	0.018	0.021	0.024
250	---	10.0	C	0.018	0.021	0.024
251	334	12.5	A	0.012	0.013	0.015
252	335	12.5	B	0.012	0.013	0.015
253	---	12.5	C	0.012	0.013	0.015
254	336	15.0	A	0.009	0.009	0.010
255	337	15.0	B	0.009	0.009	0.010
256	---	15.0	C	0.009	0.009	0.010
257	338	20.0	A	0.006	0.006	0.007
258	339	20.0	B	0.006	0.006	0.007
259	---	20.0	C	0.006	0.006	0.007

1/ The corresponding maximum wattage losses, which in no case shall exceed 3.0 watts, may be calculated as I^2R or I^2Z.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.

TABLE II. Cross reference - superseded part numbers.

Superseded MIL-C-39019/4B part number M39019/4-	New part number M39019/04-	Superseded MIL-C-39019/4B part number M39019/4-	New part number M39019/04-	Superseded MIL-C-39019/4B part number M39019/4-	New part number M39019/04-
1	200	42	231	83	218
2	201	43	230	84	219
3	200	44	231	85	218
4	201	45	230	86	219
5	200	46	231	87	227
6	201	47	230	88	228
7	200	48	231	89	227
8	201	49	230	90	228
9	200	50	231	91	227
10	201	51	248	92	228
11	209	52	249	93	233
12	210	53	248	94	234
13	209	54	249	95	233
14	210	55	248	96	234
15	209	56	249	97	233
16	210	57	248	98	234
17	209	58	249	99	236
18	210	59	248	100	237
19	209	60	249	101	236
20	210	61	254	102	237
21	215	62	255	103	236
22	216	63	254	104	237
23	215	64	255	105	242
24	216	65	254	106	243
25	215	66	255	107	242
26	216	67	254	108	243
27	215	68	255	109	242
28	216	69	254	110	243
29	215	70	255	111	245
30	216	71	257	112	246
31	224	72	258	113	245
32	225	73	257	114	246
33	224	74	258	115	245
34	225	75	257	116	246
35	224	76	258	117	251
36	225	77	257	118	252
37	224	78	258	119	251
38	225	79	257	120	252
39	224	80	258	121	251
40	225	81	218	122	252
41	230	82	219		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV, AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5925-0138-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:08Z
Check the source to verify that this is the current version before use.